

第5讲. 复合函数微分. 方向导数. 梯度.

作业: P44. 2(2). 4). 11(3).

P50. 2(2). 4(3).

 $f(x, y)$. P_0 . $f'_x(x, y)$. $f'_y(x, y)$. 在 P_0 连续. 则 f 在 P_0 可微.
(连续) $C^k(D)$: D 上 k 阶偏导函数连续的全称.
又称 k 阶连续可微.

一. 复合函数微分.

定理1. 设 $z = f(x, y)$ 可微. $x = x(t)$. $y = y(t)$ 可导. 则

$$z(t) = f(x(t), y(t)) \text{ 可导. 且 } \frac{dz}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}.$$

证: $\forall t$. Δt . x . Δx y . Δy $\therefore f(x, y)$ 在 (x, y) 可微.

$$\frac{\Delta z}{\Delta t} = f'_x(x, y) \underbrace{\frac{\Delta x}{\Delta t}}_{\downarrow x'(t)} + f'_y(x, y) \underbrace{\frac{\Delta y}{\Delta t}}_{\downarrow y'(t)} + \underbrace{\varepsilon_1 \frac{\Delta x}{\Delta t}}_{\downarrow 0} + \underbrace{\varepsilon_2 \frac{\Delta y}{\Delta t}}_{\downarrow 0}.$$

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \varepsilon_i = 0. \\ i=1, 2.$$

$$\Delta t \rightarrow 0 \Rightarrow \begin{cases} \Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \end{cases}$$

$$\frac{dz}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t} = f'_x(x, y) x'(t) + f'_y(x, y) y'(t).$$

推论1. 设 $z = f(x_1, x_2, \dots, x_n)$ 可微. $x_i = x_i(s, t)$. ($i=1, 2, \dots, n$) 存在偏导数.则 $z = f(x_1(s, t), x_2(s, t), \dots, x_n(s, t))$ 存在偏导数.

$$\text{且 } \begin{cases} \frac{\partial z}{\partial s} = \frac{\partial f}{\partial x_1} \cdot \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \cdot \frac{\partial x_2}{\partial s} + \dots + \frac{\partial f}{\partial x_n} \cdot \frac{\partial x_n}{\partial s} \\ \frac{\partial z}{\partial t} = \frac{\partial f}{\partial x_1} \cdot \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \cdot \frac{\partial x_2}{\partial t} + \dots + \frac{\partial f}{\partial x_n} \cdot \frac{\partial x_n}{\partial t} \end{cases}$$

$$\begin{cases} \frac{\partial z}{\partial t} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \cdot \frac{\partial x_i}{\partial t} \end{cases}$$

例1. 设 $u = F(x, y, z)$ 可微. $z = f(x, y)$ 可微. $y = \varphi(x)$ 可导.

$$\text{则 } \frac{du}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \cdot \varphi'(x) + \frac{\partial F}{\partial z} \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \varphi'(x) \right)$$

$$u(x) = F(x, \varphi(x), f(x, \varphi(x))) \text{ 可导.}$$

$$z = f(x, \varphi(x))$$

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \varphi'(x)$$

例2. 设 $D = \{(x, y) \mid x^2 + y^2 = 1\}$. $f(x, y) \in C(D)$. 满足 $f(0, 1) = f(1, 0)$

证明: 至少存在两点满足 $y f'_x(x, y) = x f'_y(x, y)$
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证明:

$$\text{令 } \begin{cases} x = \cos \theta \\ y = \sin \theta \end{cases} \quad \theta \in [0, 2\pi].$$

$$\text{则 } \varphi(\theta) = f(\cos \theta, \sin \theta) \text{ 在 } (0, 2\pi) \text{ 内可导}$$

$[0, 2\pi]$ 连续.

$$\varphi(\frac{\pi}{2}) = f(0, 1)$$

||

$$\varphi(0) = f(1, 0) = \varphi(2\pi)$$

$$\exists \theta_1 \in (0, \frac{\pi}{2}), \exists \theta_2 \in (\frac{\pi}{2}, 2\pi)$$

$$\varphi'(\theta_1) = 0, \quad \varphi'(\theta_2) = 0$$

$$\varphi'(\theta) = f'_x(\cos \theta, \sin \theta) \cdot (-\sin \theta) + f'_y(\cos \theta, \sin \theta) \cdot \cos \theta$$

$$= -y f'_x(x, y) + x f'_y(x, y)$$

$$(x_i, y_i) = (\cos \theta_i, \sin \theta_i) \in D, \quad i=1, 2. \quad \text{满足 } -y_i f'_x(x_i, y_i) + x_i f'_y(x_i, y_i) = 0.$$

2. 一阶全微分形式不变性.

$$z = f(u, v) \in C^1, \quad u = u(x, y) \in C^1, \quad v = v(x, y) \in C^1.$$

$$z = z(x, y) = f(u(x, y), v(x, y))$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y}$$

$$dz = \left(\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} \right) dx + \left(\frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial y} \right) dy$$

$$= \frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + \frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right)$$

$\underbrace{\hspace{10em}}_{du} \qquad \qquad \qquad \underbrace{\hspace{10em}}_{dv}$

$$\therefore dz = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv.$$

例3. 设 $x = e^u \cos v$, $y = e^u \sin v$, $z = uv$. 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

解:

$$z = uv$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$dz = u dv + v du$$

$$dx = e^u \cos v du - e^u \sin v dv$$

$$dy = e^u \sin v du + e^u \cos v dv$$

$$du = e^{-u} (\cos v dx + \sin v dy)$$

$$dv = e^{-u} (-\sin v dx + \cos v dy)$$

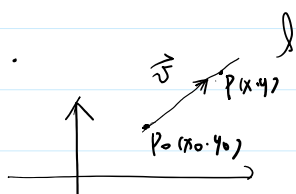
$$\therefore dz = u e^{-u} (\cos v dy - \sin v dx) + v e^{-u} (\cos v dx + \sin v dy)$$

$$= e^{-u} (v \cos v - u \sin v) dx + e^{-u} (u \cos v + v \sin v) dy$$

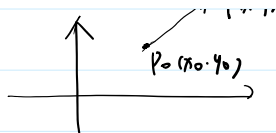
$$\therefore \frac{\partial z}{\partial x} = e^{-u} (v \cos v - u \sin v)$$

$$\frac{\partial z}{\partial y} = e^{-u} (u \cos v + v \sin v)$$

3. 方向导数.



$$\vec{r} = (v_1, v_2).$$



$$\vec{v} = (v_1, v_2).$$

$$\text{设 } \|\vec{v}\| = 1$$

$$\overrightarrow{P_0P} = t \vec{v} \quad t \in \mathbb{R}.$$

$$\begin{cases} x = x_0 + tv_1 \\ y = y_0 + tv_2 \end{cases} \quad t \in \mathbb{R}. \quad (x, y) \in l.$$

$$\|\overrightarrow{P_0P}\| = |t|.$$

射线 l : 以 P_0 为起点. $\vec{v}$ 方向.

$$\begin{cases} x = x_0 + tv_1 \\ y = y_0 + tv_2 \end{cases} \quad (t \geq 0)$$

定义1. 设 $z = f(x, y)$ 在 $P_0(x_0, y_0)$ 邻域内有定义. l 以 P_0 为起点. $\vec{v}$ 为方向单位向量. 当 $P(x, y)$ 沿射线 l 趋近 P_0 .

$$\text{若 } \lim_{t \rightarrow 0^+} \frac{f(x_0 + tv_1, y_0 + tv_2) - f(x_0, y_0)}{t} \text{ 存在}$$

则称 $z = f(x, y)$ 在 P_0 沿 $\vec{v}$ 方向导数. 记为 $\frac{\partial f}{\partial \vec{v}}|_{P_0}$. 或 $\frac{\partial f}{\partial \vec{v}}(P_0)$. 或 $f'_{\vec{v}}(P_0)$.

注: ①. t 沿 $\vec{v}$ 有向距离.

②. $\frac{\partial f}{\partial \vec{v}}|_{P_0} > 0$ 则 $f(x, y)$ 在 P_0 沿 $\vec{v}$ 方向值增加.

③. 若 $\frac{\partial f}{\partial x}|_{P_0}$ 存在. 则 $\frac{\partial f}{\partial \vec{i}}|_{P_0} = \frac{\partial f}{\partial x}|_{P_0}$. 且 $\frac{\partial f}{\partial (-\vec{i})}|_{P_0} = -\frac{\partial f}{\partial x}|_{P_0}$.

$$\frac{\partial f}{\partial x}|_{P_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} \text{ 存在.}$$

$$\frac{\partial f}{\partial \vec{i}}|_{P_0} = \lim_{t \rightarrow 0^+} \frac{f(x_0 + t, y_0) - f(x_0, y_0)}{t} = \frac{\partial f}{\partial x}|_{P_0}.$$

$$\vec{i} = (1, 0) \quad -\vec{i} = (-1, 0)$$

$$\frac{\partial f}{\partial (-\vec{i})}|_{P_0} = -\lim_{t \rightarrow 0} \frac{f(x_0 - t, y_0) - f(x_0, y_0)}{-t} = -\frac{\partial f}{\partial x}|_{P_0}.$$

例4. 求 $f(x, y) = \begin{cases} 1 & xy \neq 0 \\ 0 & xy = 0 \end{cases}$ 在 $(0, 0)$ 偏导数. 设 $\vec{v} = (v_1, v_2)$ ($v_1, v_2 \neq 0$) 方向.

$$\text{解: } \frac{\partial f}{\partial x}|_{(0,0)} = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0$$

$$\frac{\partial f}{\partial y} \Big|_{(0,0)} = 0.$$

$$\frac{\partial f}{\partial \vec{v}} \Big|_{(0,0)} = \lim_{t \rightarrow 0^+} \frac{f(t v_1, t v_2) - f(0,0)}{t} = \lim_{t \rightarrow 0^+} \frac{1}{t} = +\infty.$$

例 1. $f(x,y) = \sqrt{x^2+y^2}$. $(0,0)$.

2. $f(x,y) = \begin{cases} \frac{x^2 y}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ 在 $(0,0)$

偏导数存在

连续

可微

沿任意方向方向导数.

4. 梯度.

设 $z = f(x,y)$ 在 p_0 可微. 则 $\text{grad} f(p_0) = (f'_x(p_0), f'_y(p_0))$ 称为 $f(x,y)$ 在 p_0 梯度
 $\parallel$
 $\nabla f(p_0)$

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right)$$

$$\vec{v} = (v_1, v_2), \quad \|\vec{v}\| = 1.$$

$$\frac{\partial f}{\partial \vec{v}} \Big|_{p_0} = \lim_{t \rightarrow 0} \frac{f(x_0 + t v_1, y_0 + t v_2) - f(x_0, y_0)}{t} = z'(t)$$

$$z(t) = f(x_0 + t v_1, y_0 + t v_2) \quad z(t) \text{ 在 } t=0 \text{ 可导}$$